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Privacy-Preserving Outbreak Surveillance: Enabling Stronger Differential Privacy Through Geospatially-Aware Machine Learning

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January 16, 2026

A Progress Report submitted in partial fulfilment
of the requirements for the degree of
MAI (Computer Engineering)

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1 Introduction

Security and Cryptography is a domain that has been of utmost importance in the digital era and more so now with the ever evolving AI. There are outbreak surveillance systems like for the COVID-19 pandemic which involved sharing data or making data public. The main purpose of sharing the data is to enable users (in this case stakeholders responsible for taking action) to appropriately analyze the data to make inferences and plan pandemic response efforts. These data often contain privacy content of individuals which, if made public, can harm the individual directly or indirectly.

1.1 Differential Privacy Application

The data is studied as a whole by the user, where one cannot identify any individual from the data. These datasets which are made available to everyone protect the individual's identity by using differential privacy mechanism. In this mechanism, a privacy budget is maintained and noise is introduced in the data to increase the privacy of data. There is a trade-off between data privacy and data-utility. Increasing data privacy decreases data-utility because it makes the data too noisy and vise-versa.

Australia's CRISPER(COVID-19 Real-time Information System for Preparedness and Epidemic Response) was first of its kind to use differential privacy for this purpose instead of old techniques such as De-identification, Aggregation, Authentication, Authorization, and Encryption [1, 2]. The issue with this privacy utility trade-off often restricts the use of differential privacy to its full potential.

There is a gray area in this application that involves exploring false positives in the data. These have a passive effect on the utility of the data. Our research focuses on applying DP + add ML post-processing to reduce false positives, and this would enable using a smaller privacy budget while maintaining utility that CRISPER could not achieve.

2 Overall Goal and Objectives

Our main goal for this dissertation is to design and implement a geospatially-aware machine learning based post-processing to the differentially private data, hence finding evidence of room to increase privacy of the dataset whilst maintaining utility or providing better utility. There are objectives we aim to achieve in our research:-

2.1 Objectives

1. Find an evaluation parameter to compare our research with the existing differential privacy application by the 3rd week of November.
2. Experiment and decide on the threshold value for filtering out false positives before February.
   (a) by analyzing the plot of the data and finding the point of inflections.
   (b) study any patterns in the data to figure out if any kind of existing method like time-series data analysis or similar can be useful.
3. Research the weight penalty method previously applied in different applications by the 1st week of February.
   (a) if significant previous work is available, check if reuse for our case is possible.
   (b) if not, define a new weight penalizing method and justify it.
4. Train the data using different models and choose the best model to achieve our main goal by the 3rd week of February.
5. Implement this for another dataset beside COVID-19 for expansion and more detailed evaluation in the 1st week of March.

3 Literature Review

3.1 Australia's CRISPER

It is a near-real-time COVID-19 information platform that takes heterogeneous public health data, cleans it, and then displays it using interactive dashboards, maps, and automated alerts. The aim is to expand it so that it also ingests national list-line data and not just the public health data. To mitigate the re-identification attacks CRISPER aims to apply differential privacy to non-critical aggregates(e.g., demographics), while maintaining precision where required(e.g., total cases). The introduction of differential privacy aims to take a step forward in enhancement of security against modern re-identification attacks that are now vulnerable against traditional approaches such as aggregation, authentication, and authorization. Differential privacy carefully adds calibrated-noise to the data such that inclusion or exclusion of a single individual does not affect the data a lot. One of its limitation is that the balance between privacy and data utility is hard to maintain, where we have to compromise security for precision [1, 2].

3.2 Cost-sensitive Learning

In this paper, the authors explore the cost-sensitive versions of logistic regression and huberized SVM, and experiment this on both synthesized data and real healthcare datasets. Their findings reflect that these methods significantly reduce misclassifications compared to other cost-insensitive approaches. They fix the privacy budget for the training data and define the learning goal to approximately minimize total misclassification. The two variants output perturbation and objective perturbation both tested on synthetic data to study the privacy-cost trade-offs. For real-world datasets like Australian credit, German credit, large credit-card default datasets, and medical datasets like Parkinson's speech, diabetic retinopathy leads to the discovery that misclassification cost decreases with increasing noise in the data as expected [3].

3.3 Post-processing Theorem

A key property of a differential privacy mechanism is the post-processing theorem which states that any computation performed on the output of a differentially private mechanism does not make the data less private. In mathematical terms, If A is a ϵ-differentially private algorithm and a function f is applied to output A, then the resulting composite function f(A[DP]) remains ϵ-differentially private.

In the first paper, authors design a mechanism which outputs a noisy AUC(Area Under Curve)/AP(Average Precision) with a proven ϵ-DP guarantee for the test set. Using these noisy metrics for model comparisons, or for plotting symmetric bi-normal ROC curves the authors demonstrate an example of pure post-processing and discuss that no extra privacy budget is consumed in doing so. They demonstrate that once a model's output is differentially private, any post-processing (e.g., calculating evaluation metrics) on that output remains differentially private without needing extra noise or weakening privacy [4].

In the second paper, authors propose two algorithms that use DP. It basically has 2 layers, where the first layer is the data layer that shows the noise is injected and the second layer or logical layer where they solve linear programming problems and run gradient-descent style updates shows that these steps are implemented over the noisy summaries rather than the raw data highlighting the post-processing aspect of it. This framework uses the differential privacy post-processing theorem: when the initial classifier is DP, any function (including the fairness post-processing adjustment) applied to its output cannot worsen the privacy guarantee [5].

3.4 Differential Privacy in Deep Learning

In this paper, the authors give a very good example of how DP is important. For instance, if Alice's COVID-19 diagnosis was recorded in a hospital dataset after 20 other cases were already recorded, an adversary can infer Alice's diagnosis by submitting two queries to the dataset, one before and one after Alice's registration. These are called neighboring datasets and their querying may lead to accurate return values, allowing adversaries to infer sensitive information about the different records in the neighboring dataset. To prevent such attacks, DP ensures that any response from the dataset includes added noise, thereby preventing the adversary from learning the exact private information. The results suggest, noise acts as a form of regularization, reducing the model's tendency to overfit to specific data points, which leads to better generalization. Reduced overfitting helps lower false-positive predictions, meaning the model is less likely to incorrectly classify negative examples as positive [6].

4 Current Status of Work

The work has progressed at a steady rate with more time given to the research. At this point, all the objectives are clear and work is being put towards achieving those in the best possible way with no stone left unturned with the research work. For the outbreak surveillance data we have chosen the COVID-19 dataset. Since the work involves machine learning and pre-processing of the dataset, the code is written all in python programming language with all the support of different libraries that it provides.

The first step involved choosing the dataset according to our need and filtering out columns of our interest. In order to achieve what we aim, it is very important to understand the data as a whole and thus we narrowed down the data we want to focus on. Our approach involves dealing with geo-spatial data and for that we picked the *iso_code* column and sorted the data. Then we cleaned the data for numerical columns so we can remove NaN entries which gives error when doing aggregate operations on these columns. Then we define a new column called *population_weight* for normalizing the population weights. This column is used in defining the geo-spatial weight penalty function which we will use later in training our machine learning model. Defining threshold is important so that we can know the confirmed cases in the outbreak and it is still exploratory on which value to choose. Plotting the graph for the *new_cases* column will help in analyzing the trend line and find point of inflections so that we can define a variable threshold value. The reason for this being that a outbreak is never consistent, it rather has phases and thus defining a static threshold will not help our case.

Choosing F1 score as the evaluation parameter for now seemed the best option. Reason for picking this parameter above others is simple as this parameter is often used for classification, it balances precision (how many predicted values are correct) and recall (how many actual positives are found) into a single value. One of our objectives is to filter out true positives from false positives and this metric will help in best assessing how well we are able to do it. We are introducing the Laplace noise for different epsilon values for exploratory purposes.

5 Reflective & Ethical Considerations

5.1 Self Review

In this time, the progress has not been as fast as it could have been. Choosing an open ended project would always have proven to be difficult and I still chose this knowing that. Since the domain of security and cryptography is wide and ever-so complicated, it was really hard for me to narrow down to what I could do. It was not until November when I finally decided in what direction I want to lead my research toward, and then it took me a whole month to write the pseudo-code to get a feeling of how I can implement it. Regular meetings with my supervisor and discussions on Microsoft teams chat made sure that my progress is never hindered. At times I felt like I was procrastinating because I did not understand things or the lack of motivation due to other academic work. This must have been the cause of my slow progress, but with the appropriate guidance from my supervisor I did not go off track completely. I have not progressed much in the last month because of exams and winter holidays but now in the remaining weeks I plan to accomplish my main goal.

5.2 Ethics or Gender Dimension

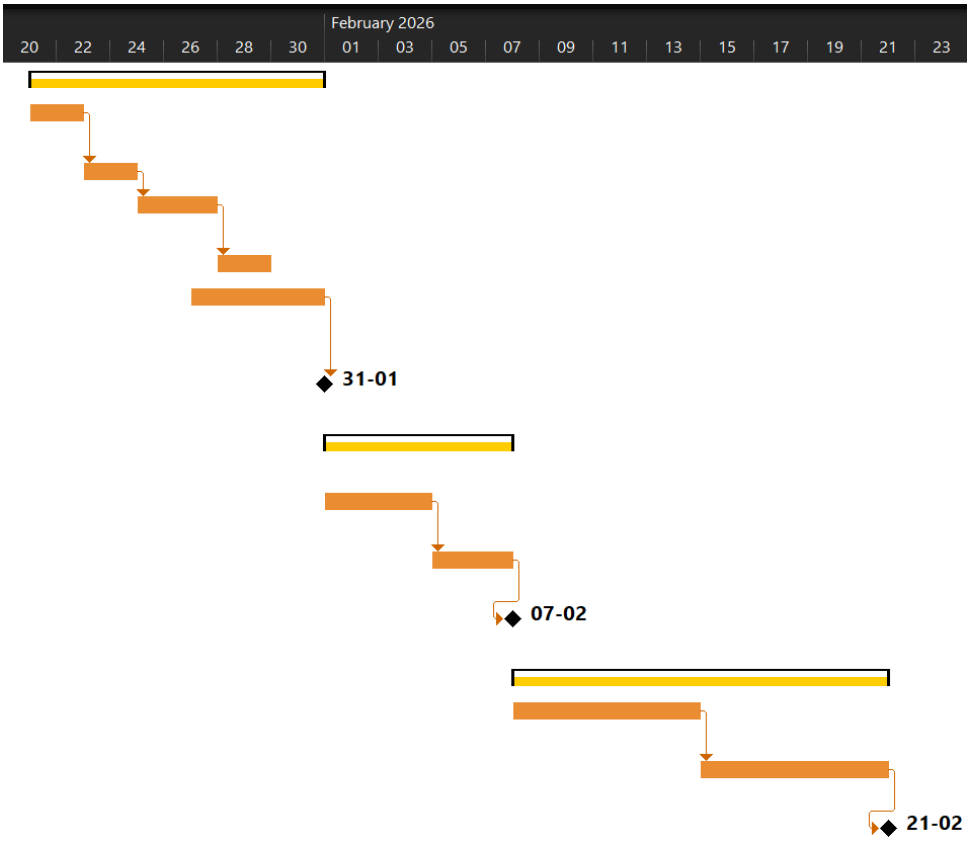
With the ethical considerations in mind, we have picked up the COVID-19 dataset that was public and free to use from Kaggle, which is provided by "Our World in Data". The data on this site, as mentioned comes mostly from third-party providers (such as WHO, UN, and World Bank) and citing them is important. Also some of the data and charts exclusively published by "Our World in Data" is also free to use with appropriate citing of it and the logo on any charts used. We will adhere to this and make sure to cite properly all the resources used in our research work. We do not have any gender dimension to our research work as we interpret the data as a whole. Our main goal of this dissertation aligning and thus we do not single out any individuals.

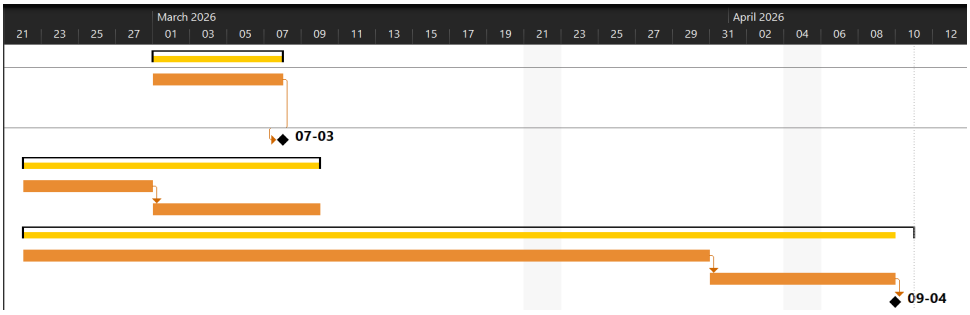
6 Project Management

6.1 Gantt Chart

	Task Mode	Task Name	Duration	Start	Finish	Predecessors
1		**Finding threshold value**	**11 days**	**Wed 21-01-26**	**Sat 31-01-26**	
2		Plot graph for new cases	2 days	Wed 21-01-26	Thu 22-01-26	
3		Analyze graph	2 days	Fri 23-01-26	Sat 24-01-26	2
4		Research time series data	3 days	Sun 25-01-26	Tue 27-01-26	3
5		Define threshold value	2 days	Wed 28-01-26	Thu 29-01-26	4
6		Research on how to define variable threshold value	5 days	Tue 27-01-26	Sat 31-01-26	
7		Finalize threshold value	0 days	Sat 31-01-26	Sat 31-01-26	6
8		**Finding weight penalty function**	**7 days**	**Sun 01-02-26**	**Sat 07-02-26**	
9		Research on existing implementations	4 days	Sun 01-02-26	Wed 04-02-26	
10		Define weight penalty functon	3 days	Thu 05-02-26	Sat 07-02-26	9
11		Finalize weight penalty function	0 days	Sat 07-02-26	Sat 07-02-26	10

	Task Mode	Task Name	Duration	Start	Finish	Predecessors
12		**Model experimentation**	**14 days**	**Sun 08-02-26**	**Sat 21-02-26**	
13		Research models commonly used	7 days	Sun 08-02-26	Sat 14-02-26	
14		Experiment with different models	7 days	Sun 15-02-26	Sat 21-02-26	13
15		Finalize model	0 days	Sat 21-02-26	Sat 21-02-26	14
16		**Expansion**	**7 days**	**Sun 01-03-26**	**Sat 07-03-26**	
17		Explore and implement on another dataset	7 days	Sun 01-03-26	Sat 07-03-26	
18		Note results	0 days	Sat 07-03-26	Sat 07-03-26	17
19		**Demonstration**	**15 days**	**Sun 22-02-26**	**Mon 09-03-26**	
20		Prepare Presentation	7 days	Sun 22-02-26	Sat 28-02-26	
21		Practice	8 days	Sun 01-03-26	Mon 09-03-26	20
22		**Report**	**39 days?**	**Sun 22-02-26**	**Fri 10-04-26**	
23		Draft 1	30 days	Sun 22-02-26	Mon 30-03-26	
24		Draft 2	8 days	Tue 31-03-26	Thu 09-04-26	23
25		Final Draft	0 days	Thu 09-04-26	Thu 09-04-26	24





6.2 Risk Management

Risk Title	Description	Likelihood (1-5)	Impact (1-5)	Score (L*I)	Mitigation/s
Insufficient data on thresholds	Insufficient inference from the plot about trend of new cases in the outbreak, thereby impacting the decision making for deciding a threshold value .	2	4	8	Find an alternative way to dynamically define threshold values to filter out false positives from true positives. Find a new dataset with confirmed values to rule out the need of threshold value to filter.
Insufficient data on cost-sensitive learning	Insufficient data on existing cost-sensitive learning approaches may affect the definition of weight-penalty function.	4	4	16	Define a new weight-penalty function and give proper justification of the choices so that other can build upon this work in future.
Implementation Problems (Reduction of False Positives)	Software implementation of weight-penalty function in the ML model makes the utility worse for the same epsilon value.	1	3	3	A minor decrease in utility of data with significant increase in the privacy of data is acceptable.

Bibliography

[1] E. Field, A. Dyda, M. Hewett, H. Weng, J. Shi, S. Curtis, C. Law, L. McHugh, M. Sheel, J. Moore, L. Furuya-Kanamori, P. Pillai, P. Konings, M. Purcell, N. Stocks, G. Williams, and C. L. Lau, "Development of the covid-19 real-time information system for preparedness and epidemic response (crisper), australia," *Frontiers in Public Health*, vol. 9, p. 753493, 2021.

[2] A. Dyda, M. Purcell, S. Curtis, E. Field, P. Pillai, K. Ricardo, H. Weng, J. C. Moore, M. Hewett, G. Williams, and C. L. Lau, "Differential privacy for public health data: An innovative tool to optimize information sharing while protecting data confidentiality," *Patterns*, vol. 2, no. 12, p. 100366, 2021.

[3] Y. Yang, S. Huang, W. Huang, and X. Chang, "Privacy-preserving cost-sensitive learning," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 32, no. 5, pp. 2105–2116, 2021.

[4] K. Boyd, E. Lantz, and D. Page, "Differential privacy for classifier evaluation," in *Proceedings of the 8th ACM Workshop on Artificial Intelligence and Security*, ser. AISec '15. New York, NY, USA: Association for Computing Machinery, 2015, p. 15–23. [Online]. Available: https://doi.org/10.1145/2808769.2808775

[5] M. Jagielski, M. Kearns, J. Mao, A. Oprea, A. Roth, S. Sharifi-Malvajerdi, and J. Ullman, "Differentially private fair learning," *ArXiv*, vol. abs/1812.02696, 2018. [Online]. Available: https://api.semanticscholar.org/CorpusID:54447781

[6] Y. Wang, Q. Wang, L. Zhao, and C. Wang, "Differential privacy in deep learning: Privacy and beyond," *Future Gener. Comput. Syst.*, vol. 148, no. C, p. 408–424, Nov. 2023. [Online]. Available: https://doi.org/10.1016/j.future.2023.06.010